

PHASE 3 — ADVANCED BC ENGINEERING (SENIOR LEVEL)

W1D1 → W36D6 (36 Weeks)

CORE PROJECT:

“Enterprise University Platform (Production BC System)”

You will evolve your Phase 2 integration system into a:

- multi-company capable system
 - background-processing engine
 - DevOps-ready extension
 - production-grade architecture
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PHASE 3 STRUCTURE

Block	Focus
W1–W6	Background processing & async architecture
W7–W12	Multi-company & enterprise architecture
W13–W18	Upgrade handling & AppSource readiness
W19–W24	Performance engineering
W25–W30	DevOps & CI/CD
W31–W36	Final enterprise system + senior validation

BLOCK 1 — W1D1 → W6D6

BACKGROUND PROCESSING & ASYNC ARCHITECTURE

W1D1 — Job Queue Fundamentals

Theory

- Job Queue Entries
- Scheduled processing model

Build

- Create job queue setup for integration sync
-

W1D2 — Background Codeunits

Build

- Move integration logic into background execution codeunit
-

W1D3 — Async Execution Patterns

Build

- Simulate async student sync process
-

W1D4 — Error Isolation

Build

- Ensure failures do not crash job queue
-

W1D5 — Retry in Background Jobs

Build

- Add retry logic for background tasks
-

W1D6 — BLOCK TEST 1 (GATE 1)

TEST

- Run scheduled job
 - Confirm async execution works
 - Confirm retry works
-

BLOCK 2 — W7D1 → W12D6

MULTI-COMPANY & ENTERPRISE ARCHITECTURE

W7D1 — Company Model in BC

Theory

- Company isolation concepts
-

W7D2 — Cross-Company Data Handling

Build

- Read/write across companies
-

W7D3 — Config-Driven Architecture

Build

- Setup table controlling behavior per company
-

W7D4 — Data Isolation Rules

Build

- Ensure no cross-company leakage bugs
-

W7D5 — Shared Services Layer

Build

- Create shared service codeunit for all companies
-

W7D6 — BLOCK TEST 2 (GATE 2)

TEST

- Run same logic in multiple companies
 - Validate isolation works
-

BLOCK 3 — W13D1 → W18D6

UPGRADE HANDLING & APPSOURCE READINESS

W13D1 — Upgrade Codeunits

Theory

- Data upgrade lifecycle
-

W13D2 — Versioning Strategy

Build

- Define version-safe architecture
-

W13D3 — Schema Change Handling

Build

- Add safe field modifications
-

W13D4 — Data Migration Logic

Build

- Migrate old student records safely
-

W13D5 — Backward Compatibility

Build

- Ensure old APIs still work
-

W13D6 — BLOCK TEST 3 (GATE 3)

TEST

- Simulate upgrade
 - Confirm no data loss
-

BLOCK 4 — W19D1 → W24D6

PERFORMANCE ENGINEERING

W19D1 — Query Optimization

Build

- Replace inefficient loops with Set-based logic
-

W19D2 — SetLoadFields Optimization

Build

- Reduce data load overhead
-

W19D3 — Locking & Concurrency

Build

- Avoid record locking issues
-

W19D4 — Large Dataset Handling

Build

- Process bulk records efficiently
-

W19D5 — Performance Benchmarking

Build

- Measure execution time improvements
-

W19D6 — BLOCK TEST 4 (GATE 4)

TEST

- Process 10,000+ records efficiently
 - Confirm no timeout issues
-

BLOCK 5 — W25D1 → W30D6

DEVOPS & CI/CD PIPELINES

W25D1 — Git Fundamentals for BC

Theory

- Branching strategies

W25D2 — Repo Structure

Build

- Organize AL solution into CI-friendly structure

W25D3 — Build Pipelines

Build

- Automate extension build

W25D4 — Deployment Automation

Build

- Auto deploy to sandbox

W25D5 — Version Tagging

Build

- Semantic versioning strategy

W25D6 — BLOCK TEST 5 (GATE 5)

TEST

- Push code → auto build → auto deploy
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BLOCK 6 — W31D1 → W36D6

FINAL ENTERPRISE SYSTEM & SENIOR VALIDATION

W31D1 — System Architecture Review

Build

- Refactor entire solution
-

W31D2 — Modularization

Build

- Split into service layers
-

W31D3 — Observability Layer

Build

- Add monitoring logs
-

W31D4 — Final Integration Hardening

Build

- Stabilize all integrations
-

W31D5 — Full System Simulation

Build

- End-to-end testing scenario

W31D6 — FINAL PHASE 3 TEST (GATE 6)

FINAL SENIOR TEST REQUIREMENTS

You **MUST** demonstrate:

Architecture

- Clean layered system
- Modular design
- Reusable services

Integration

- Stable external API flows
- Background processing
- Retry + logging system

Enterprise readiness

- Multi-company support
- Upgrade-safe design
- Performance optimization

DevOps

- CI/CD pipeline working
- Automated deployment

PHASE 3 EXIT CRITERIA

You are ONLY considered senior-ready if:

- ✓ You design before coding
 - ✓ You build reusable systems
 - ✓ You optimize performance intentionally
 - ✓ You understand deployment pipelines
 - ✓ You can review other developers' code
 - ✓ You think in architecture, not features
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FINAL REALITY CHECK

At the end of this journey:

Phase	Identity Shift
Phase 1	AL Developer
Phase 2	Integration Engineer
Phase 3	Senior BC Engineer